

How to Read the Site-by-Month Planning Table

A field guide to `practitioner_site_month_kstar.csv`

This guide explains how to read `practitioner_site_month_kstar.csv`, the site-by-month lookup delivered with the final report. It assumes no statistical background. The report explains the method behind the numbers; this document is only about using the file.

Three things are delivered together, and they all carry the same numbers:

- `practitioner_site_month_kstar.csv` — the table itself.
- This guide — how to read it.
- The *CCS Data-Sufficiency Explorer* — the same figures on screen, one station-month at a time, with the evidence behind each one. It has its own guide, `dashboard_how_to_use.pdf`.

1 What the file contains

One row for	every station and every calendar month
Rows	2,796
Stations	233, each with 12 months
Columns	12
Calendar	2024, a leap year, so February has 29 days
Format	Plain comma-separated text, one header row, no blank cells

Each row answers one question: *how many valid days does this station need in this month before its monthly average can be reported with a useful interval?*

A fuller 166-column audit table also exists, carrying the diagnostics, provenance and simulation settings behind every number. This file is the short version for day-to-day use; the audit table stays authoritative whenever a value has to be reproduced or defended.

2 Reading a row in four steps

1. **Find the row.** Match `site` and `month`. Month is a number, 1 to 12.
2. **Check what arrived.** `operational_status` says whether the 2024 month came in whole, came in with days missing, or did not arrive at all. Section 5 sets out the three values.
3. **Read the number.** Take `recommended_k` if you do not want to choose a pattern, or the specific `k_*` column for the kind of outage you actually face.
4. **Carry `threshold_status` with it.** It says whether the number is qualified — again, Section 5.

A worked example

Station 153-0189, September:

Column	Value
site	153-0189
month	9 (September)
calendar_days	30
k_mcar	5
k_block_1	8
k_block_2	9
k_weekday_heavy	7
k_weekend_heavy	4
k_markov	6
recommended_k	9
threshold_status	certified
operational_status	incomplete_requires_prediction

September 2024 has 30 dates. This station's September did not come in whole, so the row is one the planning value is for. That value is **9 days**: the largest of the five ordinary patterns, which here is Block-2's 9. The number carries no qualification. September 2024 delivered 28 valid days at this station, comfortably past 9 — so the month is usable, even though its status reads `incomplete_requires_prediction`.

3 The six outage patterns

Each `k_*` column answers the same question under a different assumption about *which* days went missing. Where the gaps fall matters as much as how many there are.

 observed day  missing day

MCAR

Random dropouts

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

Communication glitches drop isolated days, independently of each other.

Block-1

One outage

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

A power failure keeps the counter down for one unbroken run of days.

Block-2

Two outages

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

A modem fault, then a power-supply fault days later: two separate runs.

Weekday-heavy

Working week lost

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

Weekday road-works: most of the loss lands Mon–Fri, and weekends are hit least.

Weekend-heavy

Weekends lost

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

Routine weekend maintenance disconnects the sensor every Sat–Sun.

Markov

Bursty failures

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

A loose connection: once it drops it tends to stay down, and the reverse.

Every calendar above is the *same* April, and in every one of them *exactly fourteen* days survive. A rule that only counts days sees fourteen six times over and cannot tell them apart. That is why there is a separate column for each:

Column	Pattern	Column	Pattern
k_mcar	MCAR	k_weekday_heavy	Weekday-heavy
k_block_1	Block-1	k_weekend_heavy	Weekend-heavy
k_block_2	Block-2	k_markov	Markov

These are deliberate stress tests, not a model of how equipment actually fails. They let a recommendation be tied to a stated assumption instead of an average over unknown failure modes.

4 Why weekend-heavy is reported but not recommended

recommended_k is the largest of the *five* ordinary patterns. Weekend-heavy is deliberately left out of it.

- A month holds only eight or nine weekend dates. Losing them removes a whole block of the weekly rhythm at once, and the required day count jumps.

- Weekend-heavy sets the largest of the six values in roughly **three station-months in five**, sometimes by as much as **twelve days**.
- An outage landing squarely and only on weekends is rare. Sizing every month for it would produce targets nobody can meet.

Nothing is hidden. The value is still there in `k_weekend_heavy`. If a station genuinely faces weekend risk — a recurring weekend closure, or planned weekend maintenance — read that column directly. Taking the maximum across all six `k_*` columns recovers the report’s all-pattern value.

5 The two status columns

The last two columns of the file are words rather than numbers, and they answer two different questions. `threshold_status` is about the *recommendation*: how much weight it carries. `operational_status` is about the *2024 record*: what actually arrived. Neither one is a pass or a fail, and the two are independent — every combination of them occurs in the file.

`threshold_status` — how the recommendation should be read

Every value begins with `certified`, because all 2,796 rows in this delivery carry a certified number. A word after it is a qualification on how to read that number, not a warning that it cannot be used.

Value	Rows	What it means
<code>certified</code>	2,531	No qualification. Read the number as it stands.
<code>certified_calendar_sensitivity</code>	233	Every February row, and only those. The earlier years used to work the number out had a 28-day February, while the 2024 target February has 29, so the day count had to be carried between calendars of different length. The report treats February as a sensitivity result rather than a like-for-like count.
<code>certified_finite_cohorts</code>	32	A finite answer exists, but the pool of comparable stations behind it was smaller than preferred: at least nine in each of the two roles, fewer than nineteen in at least one of them. Use the number, and treat it as resting on a thinner comparison set.

A blank number would mean the value was not certified, and `threshold_status` would say so. It is not a zero. No row in this delivery is blank.

`operational_status` — what arrived in 2024

This column describes the delivered record, not the recommendation. It never changes the planning value: a month that came in whole still carries its number, because that number is guidance for the same month in a future year.

Value	Rows	What it means
<code>complete_exact</code>	2,059	Every date in the month produced a valid whole-facility daily total. The monthly average is the plain mean of those days and is reported exactly. No prediction is involved.
<code>incomplete_requires_prediction</code>	506	The month arrived, but at least one date produced no valid total — equipment down, or the day rejected in quality control. This is the case the planning value is for: compare the month’s valid-day count against it.
<code>no_target_record</code>	231	No usable 2024 record for that station and month at all. There is nothing to assess, and the number stands as guidance for a future month.

`incomplete_requires_prediction` describes the data, not a verdict on the month. It says only that not every date produced a valid daily total. Whether the month is still usable is answered by comparing its valid-day count with the planning value, and a month flagged this way will often clear that value with room to spare — as the worked example above does.

6 What the number does and does not license

It is guidance for a future month of the same name. It is not a verdict on a month already in hand, and it does not by itself produce interval endpoints.

A published value does *not* carry over when:

- the later month has a different weekday layout, or February changes length;
- the record already holds more valid days than the recommendation;
- the pattern of missing days is already known, rather than assumed.

Issuing an actual interval for a partial month needs calibration at that exact day count under a declared pattern. Reducing a longer record down to an already-calibrated count is possible, but only if the reduction rule never looks at the traffic values *and* produces the same distribution of missing days as the pattern that was calibrated. Neither condition implies the other: a rule can ignore the data entirely and still leave the wrong shape of gaps.

The interval guarantee in the report also rests on an assumption that this station’s errors behave like those of its comparison group.

7 Column reference

The rows below are in the order the columns appear in the file.

Table 1: Data dictionary for the 12-field compact practitioner lookup, in file order. An observed-day threshold counts valid days, not missing days.

Category	Field	Type or units	Meaning and interpretation
Identity	<code>site</code>	Text	GDOT continuous-count-station identifier; one physical facility.
Identity	<code>month</code>	Integer, 1–12	Calendar-month number, January through December.
Target month	<code>calendar_days</code>	Integer days	Number of dates in the 2024 target month: 29, 30, or 31. February is 29 because 2024 is a leap year.
Planning value	<code>k_mcar</code>	Integer observed days	LCB-certified persistent-tail value under uniform fixed-size random observation.
Planning value	<code>k_block_1</code>	Integer observed days	LCB-certified value when the missing dates form one contiguous run with a uniformly selected feasible start.
Planning value	<code>k_block_2</code>	Integer observed days	LCB-certified value for two separated missing runs whose lengths differ by at most one day.
Planning value	<code>k_weekday_heavy</code>	Integer observed days	LCB-certified value when approximately 80% of deletions are assigned to weekdays, subject to capacity.
Planning value	<code>k_weekend_heavy</code>	Integer observed days	LCB-certified value when approximately 80% of deletions are assigned to weekends, subject to capacity.
Planning value	<code>k_markov</code>	Integer observed days	LCB-certified value under the conditional Markov law with symmetric stay probability 0.70.
Planning value	<code>recommended_k</code>	Integer observed days	The recommended target: the largest of the five ordinary patterns, defined only for a complete numerical panel. Weekend-heavy is deliberately excluded; see Section 4.
Status	<code>threshold_status</code>	Categorical	How the recommendation should be read: <code>certified</code> , <code>certified_calendar_sensitivity</code> , or <code>certified_finite_cohorts</code> . Set out in Section 5.
Status	<code>operational_status</code>	Categorical	What the 2024 record held: <code>complete_exact</code> , <code>incomplete_requires_prediction</code> , or <code>no_target_record</code> . Set out in Section 5.